KAIZEN MATHS

Lesson starter

Straight Line Graphs and Gradients

Finding Gradient

Name _____

S1

Find the gradient of the line passing through (-6, 0) and (-8, -5)

S2

Find the gradient of the line: $2x + 4y + 4 = 0$

S3

Find the gradient of the line passing through (0, 3) and (4, 3)

S4

Find the gradient of the line: $2x + 3y + 2 = 0$

Show your working

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ALG-FCA4-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Straight Line Graphs and Gradients

Finding Gradient

Name

E1

Find the gradient of the line passing through (-1, -8) and (-4, 4)

E2

Find the gradient of the line: 6x + 5y + 1 = 0

E3

Find the gradient of the line passing through (-2, 1) and (2, 0)

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ALG-FCA4-L1 · E

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ALG-FCA4-L1 · E

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ALG-FCA4-L1 · E

Show your working

Teacher answers & working

Straight Line Graphs and Gradients · Finding Gradient

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S1

Find the gradient of the line passing through (-6, 0) and (-8, -5)

Answer:

$m = \frac{5}{2}$

Working:

Use the gradient formula: m = $\frac{y_2-y_1}{x_2-x_1}$

Point 1: (-6, 0), point 2: (-8, -5)

$m = \frac{-5-(0)}{-8-(-6)}$

$m = \frac{-5}{-2}$

$m = \frac{5}{2}$

S2

Find the gradient of the line: 2x + 4y + 4 = 0

Answer:

$m = \frac{-1}{2}$

Working:

Rearrange to y = mx + c.

$2x + 4y + 4 = 0$

$4y = -2x - 4$

$y = -\frac{1}{2}x - 1$

Therefore, gradient = $\frac{-1}{2}$

S3

Find the gradient of the line passing through (0, 3) and (4, 3)

Answer:

$m = 0$

Working:

Use the gradient formula: m = $\frac{y_2-y_1}{x_2-x_1}$

Point 1: (0, 3), point 2: (4, 3)

$m = \frac{3-(3)}{4-(0)}$

$m = \frac{0}{4}$

$m = 0$

Teacher answers & working

Straight Line Graphs and Gradients · Finding Gradient

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S4

Find the gradient of the line: 2x + 3y + 2 = 0

Answer:

$m = \frac{-2}{3}$

Working:

Rearrange to y = mx + c.

$2x + 3y + 2 = 0$

$3y = -2x - 2$

$y = -\frac{2}{3}x - \frac{2}{3}$

Therefore, gradient = $\frac{-2}{3}$

E1

Find the gradient of the line passing through (-1, -8) and (-4, 4)

Answer:

$m = -4$

Working:

Use the gradient formula: m = $\frac{y_2-y_1}{x_2-x_1}$

Point 1: (-1, -8), point 2: (-4, 4)

$m = \frac{4-(-8)}{-4-(-1)}$

$m = \frac{12}{-3}$

$m = -4$

E2

Find the gradient of the line: 6x + 5y + 1 = 0

Answer:

$m = \frac{-6}{5}$

Working:

Rearrange to y = mx + c.

$6x + 5y + 1 = 0$

$5y = -6x - 1$

$y = -\frac{6}{5}x - \frac{1}{5}$

Therefore, gradient = $\frac{-6}{5}$

Teacher answers & working

Straight Line Graphs and Gradients · Finding Gradient

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E3

Find the gradient of the line passing through (-2, 1) and (2, 0)

Answer:

$m = \frac{-1}{4}$

Working:

Use the gradient formula: $m = \frac{y_2 - y_1}{x_2 - x_1}$

Point 1: (-2, 1), point 2: (2, 0)

$m = \frac{0-(1)}{2-(-2)}$

$m = \frac{-1}{4}$

$m = \frac{-1}{4}$